Examiner only

2. (a) Define the emf of a cell. [2]

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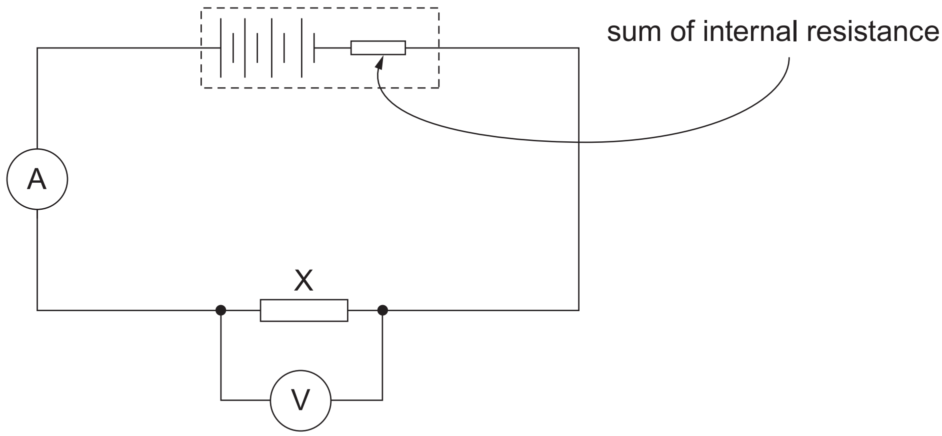
(b) Four identical cells each having an emf of 1.50 V are connected in series to form a 6.00 V battery. The battery is connected across resistor X as shown.



The reading on the ammeter is 0.30 A and the reading on the voltmeter is 5.40 V.

(i) Show that the resistance of resistor X is 18 Ω. [1]

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(ii) Determine the value of the internal resistance of **each individual cell**. [3]

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(iii) Calculate the total power dissipated in the internal resistance of the 6.00 V battery. [2]

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(iv) Resistor X is now replaced with a different resistor, Y. Resistor Y has half the resistance of resistor X. Seren states ‘as the resistance is halved, the power dissipated in the internal resistance of the battery will be twice that with resistor X.’ Evaluate Seren’s claim. [3]

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